Database Document - Addis Ababa Bus Tracking System

1. Overview

The system stores everything in one PostgreSQL database with the PostGIS extension: 15 tables, 5 enumerated types and one migration file.

Item	Value
Database	PostgreSQL 16 with PostGIS 3.4 (local Docker image `postgis/postgis:16-3.4` ; Neon online)
Access layer	Drizzle ORM in the Node/TypeScript backend
Schema source	`backend/src/db/schema/` (`fleet.ts` , `routes.ts` , `stops.ts` , `tracking.ts` , `geometry.ts`)
Migration	`backend/drizzle/0000_special_morg.sql` , applied with `drizzle-kit migrate`
Map reference system	SRID 4326 (longitude, latitude)
Local database	container `bus` , database `bus_tracker` , host port 5434

The tables fall into three groups: map data (routes and stops), fleet and people (buses, logins, shifts, warnings) and tracking (GPS points, terminal departures, learned travel times).

The migration does not create the PostGIS extension. On a new database run `CREATE EXTENSION IF NOT EXISTS postgis;` once before migrating.

2. Conventions and enumerated types

Table and column names use lower-case snake_case. Most tables use an auto-numbered integer `id` as the primary key; `location_pings` uses a 64-bit number because it grows fastest. All timestamps are

timestamptz (time zone aware). Positions are PostGIS geometry columns in SRID 4326, stored as longitude then latitude.

Enum	Values	Used by
assignment_status	assigned, on_trip, at_terminal, ended	assignments.status , assignment_changes.status_at_change
service_type	regular, premium, special	assignments.service_type
crowding_status	not_reported, seats_available, standing_only, full	assignments.crowding , location_pings.crowding , departures.crowding_at_departure
flip_type	auto, confirmed	departures.flip_type
day_type	weekday, weekend	segment_times.day_type

3. Map data tables

Four tables describe the network: which routes exist, the line each direction follows, the stops, and the order of stops on each direction. PK = primary key, FK = foreign key.

routes

One row per bus route.

Column	Type	Rules	Meaning
id	serial	PK	Route id
route_number	text	not null, unique	Number shown to riders, e.g. 49
name	text	optional	Route name in English
name_am	text	optional	Route name in Amharic

route_directions

One row per direction of a route (for example toward Ayat and toward Megenagna). Each holds the road line used for progress and ETA.

Column	Type	Rules	Meaning

id	serial	PK	Direction id
route_id	integer	not null, FK to routes, delete cascades	Parent route
direction_name	text	not null	Label such as "toward Megenagna"
path	geometry(LineString,4326)	optional	Road line of the direction

Unique together: route_id and direction_name.

stops

One row per physical stop. The two sides of a road are separate stops.

Column	Type	Rules	Meaning
id	serial	PK	Stop id
side	text	optional	Side or direction label of the stop
place_name	text	not null	Place name in English
place_name_am	text	optional	Place name in Amharic
location	geometry(Point,4326)	not null	Stop position

route_stops

The ordered list of stops on a direction.

Column	Type	Rules	Meaning
id	serial	PK	Row id

route_direction_id	integer	not null, FK to route_directions, delete cascades	Direction

Column	Type	Rules	Meaning
stop_id	integer	not null, FK to stops, delete cascades	Stop
stop_order	integer	not null	Position on the direction; 1 is the terminal

Unique together: route_direction_id and stop_order.

4. Fleet and people tables

These tables hold buses, logins, shifts and warnings. Passwords and login tokens are never stored in plain text. vehicles, drivers, admins

Table	Columns	Rules
vehicles	id, plate	plate not null, unique
drivers	id, name, username, password_hash	name, username, password_hash not null; username unique
admins	id, name, username, password_hash	same shape as drivers

A driver row is one login per bus crew. password_hash holds a bcrypt hash.

driver_sessions and admin_sessions

One row per active login. Both tables have the same shape.

Column	Type	Rules	Meaning

id	serial	PK	Row id
driver_id / admin_id	integer	not null, FK, delete cascades	Owner of the session
token_hash	text	not null, unique	Hash of the login token
Column	Type	Rules	Meaning
created_at	timestamptz	not null, default now	Login time
expires_at	timestamptz	not null	Expiry (14 hours after login)

assignments

One shift: a bus, a driver login and a route direction.

Column	Type	Rules	Meaning
id	serial	PK	Shift id
vehicle_id	integer	not null, FK to vehicles	Bus
driver_id	integer	not null, FK to drivers	Crew login
route_direction_id	integer	not null, FK to route_directions	Current direction
status	assignment_status	not null, default assigned	Shift state

service_type	service_type	not null, default regular	Service level
started_at	timestamptz	optional	Shift start
ended_at	timestamptz	optional	Shift end
last_ping_at	timestamptz	optional	Time of newest GPS point
crowding	crowding_status	not null, default not_reported	Latest crowding report
crowding_set_at	timestamptz	optional	When crowding was reported
created_at	timestamptz	not null, default now	Row creation

assignment_changes

One row each time a crew changes route direction during a shift.

Column	Type	Rules	Meaning
id	serial	PK	Row id
assignment_id	integer	not null, FK, delete cascades	Shift
from_route_direction_id	integer	not null, FK to route_directions	Old direction
to_route_direction_id	integer	not null, FK to route_directions	New direction

status_at_change	assignment_status	not null	Shift state at that moment
changed_at	timestamptz	not null, default now	Time of change

warnings

A problem the system noticed with a bus. A warning is open while resolved_at is empty.

Column	Type	Rules	Meaning
id	serial	PK	Row id
assignment_id	integer	not null, FK, delete cascades	Shift
type	text	not null	wrong_way, off_path, bad_clock, no_signal or overdue_at_terminal
severity	text	not null, default warning	Severity label
started_at	timestamptz	not null, default now	When it began
resolved_at	timestamptz	optional	When it cleared
detail	text	optional	Extra note

5. Tracking tables

These three tables hold what buses do on the road. location_pings is the largest table.

location_pings

Every GPS point a bus phone sends.

Column	Type	Rules	Meaning
id	bigserial	PK	Point id
assignment_id	integer	not null, FK, delete cascades	Shift
route_direction_id	integer	not null, FK to route_directions	Direction when sent
recorded_at	timestamptz	not null	Corrected phone time
received_at	timestamptz	not null, default now	Server receive time
location	geometry(Point,4326)	not null	Position
speed_kmh	real	optional	Speed
accuracy_m	real	optional	GPS accuracy in metres
progress	double precision	optional	Position along the route, 0 to 1
path_distance_m	real	optional	Metres from the road line; empty while stopped
crowding	crowding_status	not null, default not_reported	Crowding at that time

crowding_set_at	timestamptz	optional	When crowding was set
time_trusted	boolean	not null, default true	False when the phone clock was wrong
Column	Type	Rules	Meaning
stationary	boolean	not null, default false	True when the bus was standing

Unique together: assignment_id and recorded_at, so a resent point is stored once.

departures

One row each time a bus reaches a terminal and later leaves it.

Column	Type	Rules	Meaning
id	serial	PK	Row id
assignment_id	integer	not null, FK, delete cascades	Shift
terminal_stop_id	integer	not null, FK to stops	Terminal
from_route_direction_id	integer	not null, FK to route_directions	Direction before the turn
to_route_direction_id	integer	optional, FK to route_directions	Direction after the turn

arrived_at	timestamptz	not null	Arrival at the terminal
departing_now_at	timestamptz	optional	When the crew tapped Departing now
departed_at	timestamptz	optional	When the bus left
flip_type	flip_type	optional	auto (detected) or confirmed (crew tapped)
crowding_at_departure	crowding_status	optional	Crowding when leaving
layover_seconds	integer	optional	Time spent waiting at the terminal

segment_times

Typical time to drive one stretch between two stops, by day type and 30-minute slot. Routes that share a stretch share its row.

Column	Type	Rules	Meaning
id	serial	PK	Row id
from_stop_id	integer	not null, FK to stops, delete cascades	Start of stretch

to_stop_id	integer	not null, FK to stops, delete cascades	End of stretch
day_type	day_type	not null	weekday or weekend
time_bucket	integer	not null	30-minute slot of the day
seconds	integer	not null	Typical travel time
measured	boolean	not null, default false	False = starting guess, true = real median
sample_count	integer	not null, default 0	Real crossings behind a measured value

Unique together: from_stop_id, to_stop_id, day_type and time_bucket.

6. Relationships and deletion rules

There are 20 foreign keys. Rows that only make sense with their parent are deleted with it (cascade); links to routes, buses and drivers are protected, so a route, bus or driver with history cannot be deleted by accident.

Child column	Parent	On delete
route_directions.route_id	routes.id	cascade
route_stops.route_direction_id	route_directions.id	cascade
route_stops.stop_id	stops.id	cascade
driver_sessions.driver_id	drivers.id	cascade
Child column	Parent	On delete
admin_sessions.admin_id	admins.id	cascade

assignments.vehicle_id	vehicles.id	no action (blocked)
assignments.driver_id	drivers.id	no action (blocked)
assignments.route_direction_id	route_directions.id	no action (blocked)
assignment_changes.assignment_id	assignments.id	cascade
assignment_changes.from_route_direction_id, to_route_direction_id	route_directions.id	no action (blocked)
warnings.assignment_id	assignments.id	cascade
location_pings.assignment_id	assignments.id	cascade
location_pings.route_direction_id	route_directions.id	no action (blocked)
departures.assignment_id	assignments.id	cascade
departures.terminal_stop_id	stops.id	no action (blocked)
departures.from_route_direction_id, to_route_direction_id	route_directions.id	no action (blocked)
segment_times.from_stop_id, to_stop_id	stops.id	cascade

In short: one route has two directions; a direction has many ordered stops; a shift links one bus, one driver login and one direction; a shift owns its GPS points, warnings, direction changes and terminal departures.

7. Indexes and constraints

Besides primary keys and foreign keys, the schema defines these indexes and unique rules.

Table	Name	Kind	Purpose

stops	stops_place_name_idx	btree on place_name	Fast stop search by name
stops	stops_location_gix	GiST on location	Nearestстop and distance queries
location_pings	pings_assignment_time_idx	btree on assignment_id, recorded_at	Latest points of a shift
location_pings	pings_location_gix	GiST on location	Spatial queries on GPS points
location_pings	pings_assignment_time_uq	unique on assignment_id, recorded_at	No duplicate points
routes	routes_route_number_unique	unique	One row per route number
route_directions	route_directions_route_id_direction_name_uniqu e	unique	One row per direction name
route_stops	route_stops_route_direction_id_stop_order_uniqu e	unique	One stop per position
segment_times	segment_times_from_stop_id_to_stop_id_day_typ e_time_bucket_unique	unique	One time per stretch, day type and slot

vehicles	vehicles_plate_unique	unique	One row per plate
drivers, admins	*_username_unique	unique	Usernames cannot repeat
driver_sessions, admin_sessions	*_token_hash_unique	unique	Token hashes cannot repeat

8. Data lifecycle

Map data is loaded once; fleet and tracking data is written while the system runs.

Data	Written by	When
Routes, directions, stops, route_stops, starting segment_times	bootstrap.ts from src/seed/route49.json and route61.json	First start, only if the routes table is empty
First admin account	bootstrap.ts from ADMIN_USERNAME and ADMIN_PASSWORD (at least 8 characters)	First start; an existing admin is never reset
Vehicles, drivers, assignments	Admin app through the backend	When staff add a bus, a login or a shift
Sessions	Login endpoints	Each login; lasts 14 hours
location_pings, assignment changes, departures	Driver app through the backend	Every GPS point, turn and terminal visit

Warnings	Backend checks and the sweeper (every 60 seconds)	When a problem starts; resolved_at is set when it clears
segment_times (measured rows)	Nightly job	Replaces a guess with the median once a slot has 3 real crossings

Seed size: route 49 has 14 stops and 1248 starting segment times per direction; route 61 has 20 stops and 576 per direction.

Growth: location_pings dominates. One bus sending a point every few seconds adds thousands of rows a day, so the table needs a retention plan before a large fleet runs. No automatic deletion exists yet; old points can be removed by date once the nightly job has used them. Warnings, departures and assignments grow slowly.

9. Setup, migration and backup

1. Create the database and enable PostGIS: CREATE EXTENSION IF NOT EXISTS postgis;
2. Set DATABASE_URL for the backend. On Neon use the direct connection string without channel_binding .
3. Apply the schema with drizzle-kit migrate .
4. Run bootstrap.ts to load the routes and create the admin.
5. Start the server. The Docker image runs steps 3 to 5 automatically in dockerentrypoint.sh .

Locally, docker-compose.yml starts the database on host port 5434 (user Rahel, database bus_tracker).

Topic	Practice
Schema changes	Edit the files in src/db/schema/ , generate a new migration, apply it; never edit the applied migration
Secrets	Keep DATABASE_URL and admin credentials in .env or host settings, never in Git

Passwords and tokens	Stored only as hashes (bcrypt passwords; hashed login tokens)
Backup	Neon keeps point-in-time history on its plan; for the local database use `pg_dump` and test a restore
Reset password	If a connection string was shared, reset the database password in Neon and update the backend setting